

Maira Arshad

Computer Science Undergraduate | OOP & Web Architecture | JavaScript
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Summary

CS sophomore with OOP and layered web-architecture experience across Node.js/Express/PostgreSQL backend internships and independent Next.js/React and C# projects.

Technical Skills

Core Concepts: Object-Oriented Programming (OOP), Layered/MVC-Style Architecture, RESTful API Design, Database Schema Design, Web Architecture

Languages: JavaScript, TypeScript, C#, Python, C++, SQL

Frameworks & Libraries: Express.js, Next.js, React, Flask, FastAPI

Databases: PostgreSQL, MySQL, SQLite

Tools: Git/GitHub, Docker, Docker Compose, VS Code

Experience

Backend AI Engineering Intern

Jul 2026 – Present

FlyRank

- Build RESTful API endpoints and structured-output pipelines using Node.js and Express.js, applying a 3-tier layered architecture (routes, services, data-access) with the repository pattern.
- Design and query PostgreSQL relational schemas independently, applying database fundamentals including primary keys, constraints, and connection pooling.
- Document and test APIs using Swagger/OpenAPI, validating status codes and response bodies against defined success criteria.

Product & Systems Intern

Jun - Aug 2025

Ustadam (AI EdTech Startup)

- Led a 4-member development team through weekly sprint cycles, coordinating task allocation and progress tracking for an AI-based EdTech MVP.

Projects

Smart Emergency Response System

C# (.NET, Windows Forms), MySQL, Google Maps API

- Independently developed a full-stack desktop application using OOP principles and layered (BL/DL/UI) architecture for ambulance dispatch and routing.
- Implemented 6 core data structures and algorithms from scratch (Priority Queue, Graph, AVL Tree, Hash Table, Linked List, QuickSort) to power real-time dispatch logic.
- Built a MySQL-backed data layer with full CRUD functionality across stations, ambulances, and emergency records.

Allied Shipping Website

Next.js 15, React, TypeScript, Tailwind CSS

- Independently built a responsive, mobile-first corporate website using component-based architecture and the Next.js App Router.
- Deployed via Vercel with a continuous deployment pipeline from GitHub.

QueryMind (SmartCRM)

PostgreSQL, Neo4j, Qdrant, FastAPI, Python

- Co-designed a 5-layer web architecture (Presentation, API Gateway, AI Intelligence, Service, Database) unifying 3 database paradigms behind one interface.

Education

University of Engineering and Technology (UET) Lahore

Expected Jul 2028

BS Computer Science

Relevant Coursework: Object-Oriented Programming, Database Management Systems, Design and Analysis of Algorithms, Computer Networks

Certifications

- AI Fluency Framework & Foundations – Anthropic
- SQL (Basic & Intermediate) – HackerRank